

A Parametric Complete-Garment Sweater Pattern with knitout

Pratyay Didwania*

Teadora Gildengers*

James McCann

pratyay@cmu.edu

tgildeng@andrew.cmu.edu

jmccann@cs.cmu.edu

Carnegie Mellon University

Pittsburgh, Pennsylvania, USA

ABSTRACT

Complete garment knitting uses knitting machines to create full garments that require minimal post-processing. We demonstrate an open-source, parametric complete-garment sweater pattern using the knitout knitting output language. Our pattern is parametric in the sense that it can generate machine knitting instructions given a wide range of body measurements and design eases. It is the first open-source parametric sweater pattern. Our demo will consist of pre-knit sweaters showcasing the various degrees of freedom in our pattern; a poster discussing our design decisions and trade-offs; and a live software demo where conference attendees can enter measurements and see a visualization of the resulting knitout code.

KEYWORDS

Knitting, Machine Knitting, Sweater, Complete Garment Knitting, Parametric Pattern, Textiles

1 INTRODUCTION

Machine knitting has become increasingly interesting to the computational fabrication community in the last 10 years. Including working out a generic language for machine knitting [McCann 2017]. Despite this interest, there has been a lack of freely available, fully-worked-out patterns for (e.g.) garments, especially complete-garment patterns (that is, patterns that don't require any sewing after the knitting completes; often known by the promotional names "WholeGarment" [Shima Seiki 2023] or "Knit And Wear" [Stoll 2023]). We decided to partially address this lack by developing a parametric pattern¹ that generates knitout code to produce custom-fit sweaters based on standard tailoring measurements.

*Both authors contributed equally to this research.

¹<https://github.com/textiles-lab/parametric-sweater>

Authors' address: Pratyay Didwania, pratyay@cmu.edu; Teadora Gildengers, tgildeng@andrew.cmu.edu; James McCann, jmccann@cs.cmu.edu, Carnegie Mellon University, 5000 Forbes Ave, Pittsburgh, Pennsylvania, USA, 15213.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2018 Association for Computing Machinery.

0730-0301/2018/8-ART111 \$15.00

<https://doi.org/XXXXXXX.XXXXXXX>

2 METHODOLOGY

We began developing our pattern with an analysis of the different elements that compose sweaters. A diagram is provided, with possible options using the latest version of the codebase in color, and future developments greyed out. Through analyzing different hand-knit sweaters, we determined that bottom-up construction would align well with the takedown system in a Shima Seiki knitting machine. Other features chosen for initial implementation were a turtleneck style, which emphasized proper neck fitting and elasticity, as well as a round yoke formed with internal decreases.

Testing began with the desire to refine the basic construction of a sweater into efficient code. For example, some early tests included stretchy bindoffs for ease of wearing sweaters, as well as cast-ons and ribs for the base of the sleeves and the body. Further tests involved determining an effective way to perform increases and decreases on knit tubes, to allow for shaping of custom-fit sweaters.

After completing these samples, we moved on to small scale sweater tests (at most 3x5 inches) to validate the shape of the garment and experiment with various approaches to joining the sleeves to the body.

Once satisfied with the shape of small scale sweaters and the structure of the armhole join, we started running half-scale sweater tests fit to a half-scale mannequin to refine short-rowing used to provide bust-shaping. In addition, we included neckline shaping options and were able to more accurately test the mobility of the armhole join.

The next step was to produce full-scale tests for real human body measurements. This revealed a problem of fabric bunching near the armhole join and tightness around the shoulder and back. We resolved the fabric bunching by carefully modifying the shape of the knit tubes before and after the armhole join, and reduced tightness by adding more fabric using short-row techniques.

Once the shape of the sweater pattern was finalized, we optimized the code to reduce carrier passes, mostly through more efficient transfer planning. This reduced production time by over 30%. With our final pattern a sweater for a medium-sized adult (53cm sleeve, 50cm torso length, 93cm torso diameter) knits in around an hour on our Shima Seiki SWG091N2 running at a 0.5m/s carriage speed.

We are currently testing and refining additional style options, such as producing sweaters in garter and seed stitch in addition to stockinette.



Figure 1: Sweater custom fit to arm length, shoulder width, with improvements to be made in the shoulder and armhole join area

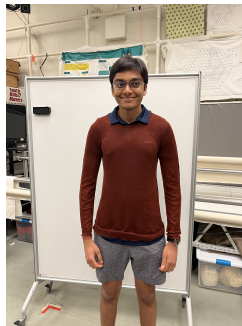


Figure 2: Final version of base sweater pattern - front

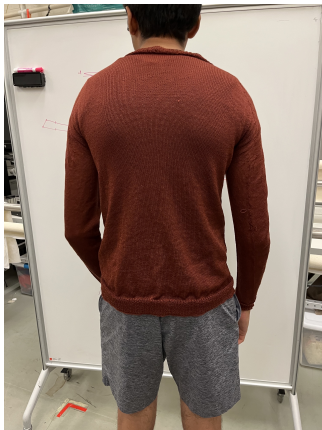


Figure 3: Final version of base sweater pattern - back

3 DEMO REQUIREMENTS

The demonstration will consist of several examples of completed sweaters, as well as stitch pattern samples, and a poster discussing



Figure 4: Artemis wears a garter stitch patterned child-size sweater.

our programming approach to various features. Conference attendees can expect to see three different sweaters that are varied in size, colour, and/or stitch pattern. We will also be ready to discuss the differences between machine knitting with knitout and hand knitting, with specific focus on adjusting size.

Our poster will depict the parts of the code that correspond to efficient knitting double-bed ribbing on a tube, short-rowing for bust shaping and shoulder adjustment, as well as neckline shaping. Each of these features will be visible in the sweaters displayed

The demonstration will also include a walkthrough of how to use the current pattern. A sample of the measurement sheet used will be provided, and we will discuss with attendees how to take measurements, enter measurements into our frontend interface, and process the measurements with our backend code to create a knitout file. The resulting knitout files will be shown using the knitout live visualizer². We will also be prepared to discuss adjusting size over multiple fittings and how to achieve this in the backend with a combination of swatching and stitch selection.

Please provide:

- Table space and a table approx 2x4 feet
- Poster space and poster stand
- Electricity to power a laptop

ACKNOWLEDGMENTS

Thanks to Lea Albaugh, Angelica Bonilla, Artemis Bonilla, Jenny Lin, Gabrielle Ohlson, and the CMU Textiles Lab for pattern ideas and knitting assistance. This material is based upon work partially supported by the National Science Foundation under Grant No. 1955444.

REFERENCES

- James McCann. 2017. The “Knitout” (.k) File Format. [Online]. Available from: <https://textiles-lab.github.io/knitout/knitout.html>.
- Shima Seiki. 2023. Wholegarment: Knit to save the earth. <https://www.shimaseiki.com/wholegarment/love/>.
- Stoll. 2023. Stoll Knit and Wear Technology. <https://www.stoll.com/en/machines/knitwear/>.

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009

²<https://textiles-lab.github.io/knitout-live-visualizer/>